Increasing awareness of local ecology to guide music composition

Xena Derks

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0. Abstract

I investigated a player's interaction with their environment as a factor in their conscious enjoyment of video game music. This investigation consisted of a review of available literature- including a discussion with an expert- and a retrospective analysis of my own work. In my conclusion I outline methods I've found through this research which might enable a musician to draw the attention of a player to their music and allow them to more consciously enjoy it- though I remain inconclusive on the question of whether this goal benefits the audience as much as it does the artist.

1. Introduction

Over the past two years, I've experienced a profound transition in my artistic and professional identity. During this period I was first introduced to musical theater, where my music became an integral part of a direct interaction with an attentive audience. As I originate from music for games, it was a notable culture shock to be put so directly into contact with an audience which I had previously been able to observe only abstractly as a target audience, if I had considered them at all.

This brought to my attention a heretofore unconsidered missing factor of artistic creation which I was, at first, unable to accurately label beyond vague phrases such as "vulnerability" or "identity." With these themes in mind I continued to engage more actively with performing arts, collaborating with trained jazz- and classical musicians and the Glassgow Royal Conservatoire in may 2023 to create Limbo, a unique and chilling exploration of guilt and self-flagellation, as well as participating in the KASKO residency, during which I performed live.

During the KASKO residency I became especially focused on a person's environment in relation to their musical perception. The entire experience centered around creating and anthology of musical-theater performances in- and around the spaces of an old church, which inevitably lead to us paying special attention to the spaces in which we would be performing. During three of my four performances the audience was positioned atypically; Suffice to say, the physical space was a prominent and recurring 'character' In these performances.

Subject

Sometime after KASKO came to an end, I had an enlightening phone call with one of the people leading the experience, Marnix Vinkenborg. Marnix helped clarify the concern I'd been developing throughout this process. Namely, he explained how composition at its core is an artform with a missing element. There's a physical, tangible context that is rarely taken into account when *writing* music, as opposed to performing it. Marnix referred to Heiner Goebbels, a composer who writes non-musical, non-sonic instructions into his scores, like instructing musicians to dodge an imaginary thunderbolt. Through this process, the physical context is brought back into the picture.

During my next project, grew in my conviction that this subject merited further investigation, and that my supportive narrative would be the perfect opportunity to perform said investigation.

Goal

I want my music to connect with my audience, and ideally I'd like to anticipate what that connection will look like during the creation process. To do so, I'm going to explore how a listener interacts with their environment (interaction with environment being shortened to 'ecology') and how I, intern, can influence that interaction to place my music in as favorable a position as possible to be experienced.

My critical review will dive into the established research of ecological musicology, reading papers and interviewing a professor of musicology with a special interest in this topic. As part of this research I'll also take a deeper dive into virtual ecology: the interaction with virtual environments, which is very important to videogame music.

After my critical review I'll be performing an experiment of my own, the methodology of which will be described in its own chapter.

Finally, I'll discuss my combined findings and what they mean for my art going forward.

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1. ecology is the study of living organisms (including humans) in relationship to their environment (Ecological society of America, 2024). This is usually a term used in biology, but professor Eric Clarke used the term to describe a certain perspective on musical perception, which is the perspective I'm adopting here. *Local* ecology designates the scope: I'm not looking at the entirety of one's cultural surroundings, but to the immediate surrounding area which might impact one's listening experience by either distracting from- or directing that experience in some way.

2. Critical review

My critical review- the first prong as outlined above- will cover three perspectives on ecological musicology:

- An introduction, as presented by the book Ways of Listening and an interview with its author, professor Eric Clarke
- An exploration of virtual ecology, with an emphasis on immersion
- An application of the first two perspectives to a concrete example: music in main-line *Mario Games*.

Through this exploration I hope to establish the necessary basis on which we can start to build in the following chapters. A lot of my critical review will be combing over scientific literature. I'm going to do my best not to make it too dense and to keep it tied to my own professional development.

2.1 Eric Clarke

I had the pleasure of reading Eric Clarke's book Ways of Listening, only to follow it up with an interview with the man himself.

2.1.1 Ways of listening

In Ways of Listening, Eric Clarke lays out the fundamentals of what he calls "the ecological approach to perception". He begins his introduction with an example:

> "Imagine that you are tidying your desk and come across an unlabeled CD. Not knowing what it is (it could be a backup disk, some student work, a CD of sound examples) you put it into your CD player and press the play button. The first track starts with a kind of blustery, rustly, crackly sound—hard to recognize to begin with—and then clearly the sound of someone eating crisps. This must be some studio project belonging to a student, you think, and curious to discover what else is on it (now that you know that it is audio and not data!) you skip to the next track—which starts with African percussion sounds over a jazzy bass line and what sounds like a sampled mbira. This is definitely someone else's CD, so you press stop, eject the CD and wonder who it belongs to and how to get it back to its rightful owner." (Clarke, 2005: 09)

What he intends to illustrate is that the experience of this sound is essentially exploratory. You question what sound you are hearing first, which might be especially disorienting, considering the initial noise of a crinkling bag amongst your messy desk, which might cause you to double check the origin.
Once you solidly determine the CD as the source, and then a bag of crisps as the source behind that, your thoughts might drift to the sound's function. Someone put this sound on this CD, presumably for some reason you're unaware of.
You then filter these musings through your cultural lens: African percussion, as a term,

is effectively meaningless if you consider the size of the continent of Afrika and its cultural variety, but in west-European countries it's a shorthand that might well pop up in your head anyway. A jazzy bass is equally evocative to someone with a musically adept ear, and a sampled mbira might bring to mind a number of (perhaps cheaply) sampled instruments thrown into an unfinished work.
Finally, you consider what action this sound requires of you. In this case, having determined that this is not your property but someone is likely still working on it, you begin to wonder how to return it to its owner.

What Clarke is effectively arguing for is a way of looking at sound as a highly structured, perceivable "object." This is in opposition to the school of thought called constructivism, which considers the world, to use an old quote also referenced by Clarke, a "blooming buzzing confusion" (James, 1892: 16) through which we derive meaning by gradually applying rules and representing a gradually more and more organized representation of the world. We hear a pitch, before we hear an interval, before we hear a chord, before we hear a structure, etc.. Clarke observes that this seems contrary to how the lay person might experience music by immediately grasping the sound of an instrument or genre, before (potentially) deconstructing what they've heard into its component elements.

What we can learn about local ecologies from Ways of Listening primarily stems from the chapter "Invariants in perception." Clarke writes:

> When perception proceeds in an unproblematic way, we are usually unaware of the sensory aspect of the stimulus information, and are only attuned to the events that are specified by stimulus structure. But when that relationship becomes problematic, the stimulus structure itself can become more evident. (Clarke, 2005: pg. 38-39)

He goes on to explain the concept of invariants, which are the basic building blocks by which we identify unstable perceptions. After all, no two rock songs are the same, but we can recognize them as rock and identify them as such by sound alone, due to a set of recognized invariants that designate them as such. However, once we recognize the invariants, it can be a point of fixation which might disallow us from perceiving further traits. Clarke demonstrates this idea using the following image:

Figure 1.1 What do you see here? (Photographer: R. C. James).

To some, this image has a distribution of shapes and forms with a great deal of nuance. However, once we spot the dalmatian and the tree, a lot of this nuance is lost as the shapes become part of distinct entities.

Once we apply all of this theory to the craft that's the subject of this paper, some puzzle pieces fall into place. Especially in interactive media, lots of different perceptual objects are merged, all with their own invariant baggage. You're looking at a screen with images, you're asked to follow instructions, you're hearing sounds that (presumably) correlate to everything you're experiencing- and on top of ALL of that, you hear music. There's no space in that context for a contemplative consideration of what you're hearing, which means your mind defaults to the invariants. You may hear rock, but have trouble to name the chords when asked at a later time.
Paradoxically, due to the lack of specific attention paid to the music there is a risk of it becoming noise, which is where the outside environment comes in. With such an overload of stimuli, you're vulnerable to outside influence as you can't consciously process them, and so you fall back on their invariants.

If you looked at the above image (without knowing what it is) while you're in a crowded mall, you might construe the shapes as being representative of entities moving around a 2D grid, displacing your environmental meaning into the media you observe.

To add onto what we've covered so far, I'd like for us to imagine an alternative scenario based on Clarke's hypothetical CD. To translate it to 2024, the CD will be transformed into a USB.

> Imagine that you aren't tidying your desk, but instead are riding the bus. You find the same USB, plug it into your laptop (cyber-security be damned) and begin to click around some of the MP3s. Where you heard a crinkling before, you now dismiss it as a snack being opened by some of the kids in front of you. The crunching potato chips, by contrast, stand out as the sound is eerily close in a context where that kind of noise is usually distant and (importantly-) visible. As you arrive at your busstop, you close your laptop and hold it under your arm as you hear the mbira sample start to play. Another man- pale, with a beanie and a goatee- gets off with you, and instead of thinking of Afrika your mind goes to hippie drum circles, weed, and a faux-intellectual discussion you had with a cousin a week ago in which you gleefully participated.

Though this example might be a bit over the top, it illustrates the central point of this paper: where your more macro ecology did not change (you're the same person with the same background and experiences, possibly in the same city) your local ecology radically shifted your interpretation of events. In a way, your invariants were moved around by virtue of the immediate environment. If you're an artist on spotify, or you make a game on a portable device, you might well fall prey to this effect- which could be frustrating if not previously considered.

2.1.2 The interview

To start this section I do need to briefly admit that I'd never expected this interview to take place. Professor Clarke's book was a fascinating read, and I was as surprised as I was thankful to hear that he was interested in speaking with me.
With that out of the way, I'm interested in talking a bit about some of the answers prof. Clarke had to the questions that came from my reading his work.

We first discussed the concept of invariants. In a highly stimulating media environment, I wanted to know whether prof. Clarke considered abstracting a piece of music an effective way of grabbing a listener's attention, because they couldn't simply fall back on (perhaps easily distorted) invariants. If so, this would reduce the need for me to manipulate a listener's ecology! I'll quote his answer verbatim, with some repetitive lines taken out for brevity (a full transcript is provided among this paper's attachments):

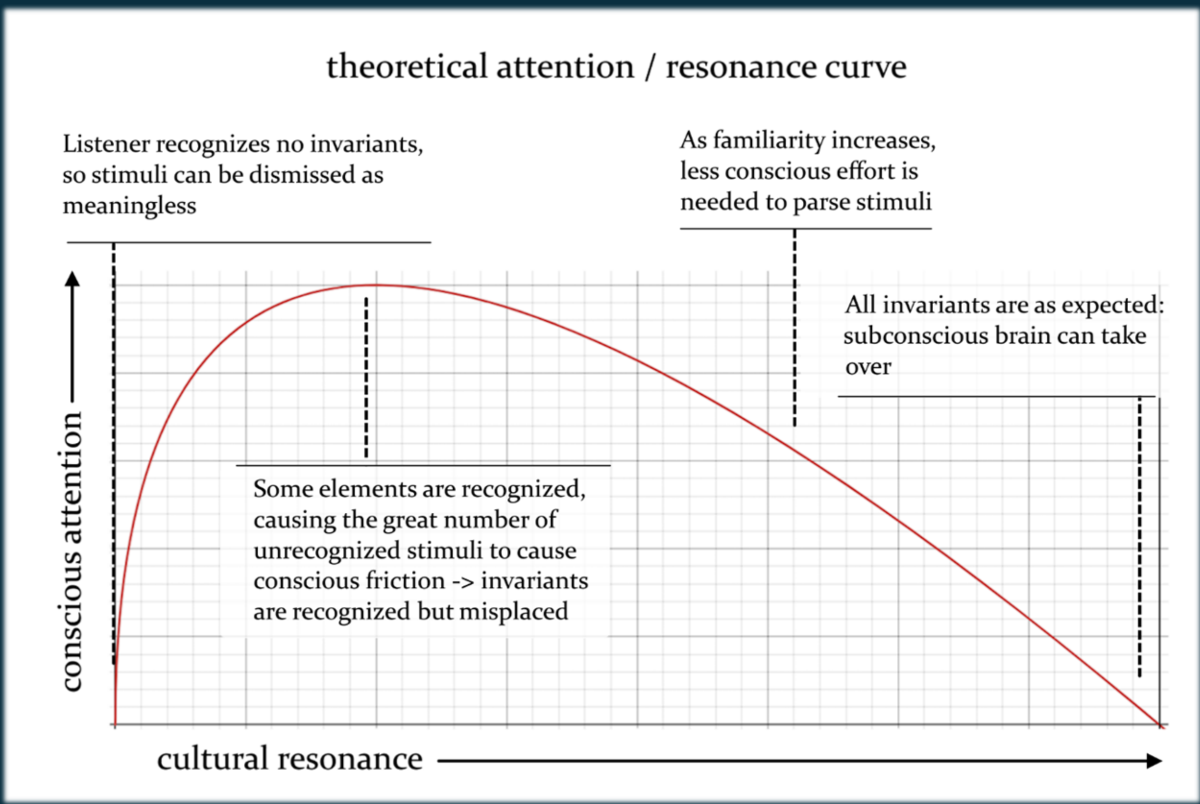
E = Eric Clarke; X = Xena Derks (myself)

"E: [...] I think that there are kind of two sides to it. One is, that if you make material more unconventional in whatever way, either by using different timbres or by using unconventional rhythmic properties or any of those kinds of attributes, in principle, yes, you expose listeners to- or you present them with material that gives them a richer and *different* understanding than they would get from being confronted with very traditional materials with which they are already very familiar. But the other side of that coin off course, if you present people with deeply unfamiliar material, then [...] they may be not well tuned to that material. In the book I use the sort of- well it's not just a metaphor, it's a reality, I used the word *tuning,* that people are *tuned* to-

X: resonating

E: exactly, and resonating with their environment. And you can only be attuned to an environment and you can only resonate with it if you have some grasp of it. If you have some kind of familiarity with it. So [...] the more unfamiliar a situation is, the more you risk people simply being bewildered and lacking in any kind of understanding of what people are confronted with." (personal correspondence, transcript enclosed)

What prof. Clarke describes a balance which one must strike between familiarity and unfamiliarity. Consider the following graph:



Here I've attempted to visualize prof. Clarke's answer. What strikes me is the similarity to the theory of the uncanny valley (Mori, 1970), which posits that, as humanoid imitations become more realistic, people's affinity for such imitations gradually increases, until a sudden drop where recognizable features are close enough- but not quite exactly- what they're *supposed* to be, where people are suddenly freaked out by what they see.

Though my proposed curve wouldn't quite fit on top of prof. Mori's (even when flipped in a way that the axes are more analogous), I still think that we could use the now decades of research into the uncanny valley as a hint towards our use of music to more consciously blend in with- or stand out from environmental stimuli. The danger is clear when we consider what we're measuring when we talk about the attention asked needed to process music: the uncanny valley graphs *affinity* in its place. As prof. Clarke described, forcing irregularity risks audience bewilderment.

A blogger, James Cassidy, made an excellent point about a possible hurdle when taking this uncanny valley approach:

> "[when creating UX] striving to achieve parity with real-life objects will always be difficult: there's only so much you can do with a mouse or touch and a two-dimensional display. These limitations effectively cap our horizontal 'realism' axis." (Cassidy, 2014)

The topic of a 'limited realism curve' is one I also discussed with prof. Clarke. I asked him about his concept of musical perception as being essentially *exploratory*, and how that applies to a medium where your influence is limited: in a real world environment you might physically turn your head to a person talking, or note the way the sound of a busker's performance becomes more and more substantial as you approach, but in a recording or in a virtual environment with a somewhat static soundtrack you would seem to lack the same kind of exploratory capacity.

Prof. Clarke agrees, to an extent, that your options are limited in this context, though he still believes some exploration is available- even to an untrained ear:

> "[...] the thing about a virtual domain, such as a piece of recorded music [...] you know, you might even try turning the volume up a bit because you hear something in the background and without deafening yourself you might be able to hear it. So there is a limited amount of actual [...] engagement with the material that might tell you a little bit more about what's going on. But you're mostly- you're absolutely right. You can't do anything to actually get into that world and find anything more out about it. But I think the equivalent kind of thing that goes on is that we can learn to train our attention to listen to different aspects of this thing, or while we listen on repeated hearing. So for instance, just a very simple thing, listening to a lot of classical music people tend to listen to, [...] the melody line, the higher line. And if you [...] listen to the bass, then suddenly you might find all kinds of things going on that you didn't notice the first time. [...] the difference of listening to the same piece with that different kind of attention or perspective can be really profound. And certainly- Im now in my late 60s and when I listen- when I go back to music I listened to when I was a teenager, especially pop music which I was particularly familiar with as a teenager, I now hear it with [...] completely different intentional tuning. I hear things and I go "oh, was that going on when I listened to it in 1975, I can't believe it" you know "I didn't hear that" or "I wasn't aware of it." It didn't *resonate* with me." (personal correspondence, transcript enclosed)

So even *without* that more physical exploration, you could still explore music if you're put into a context to do so. This might even require previous knowledge, as is emphasized by the professor's anecdote about the music of his childhood. This means that, ideally I'd have a hand in shaping that context, which is something which the next chapter will address.

2.2 Virtual Ecology

From my conversation with Eric Clarke and the research I did to elaborate on our interview, I've gotten a distinct impression that local ecology is a much more complicated topic when it comes to virtual media. In such media where you're asked to interact with- and fully immerse yourself in another world, your ecology becomes much more explicitly shaped by the work's author. A videogame world is *a world*, which means your interaction with that world constitutes its own, miniature ecology. Though you're also absolutely asked to immerse yourself in theater, you lack the agency that might cause you to internalize the world of the characters as your own.

In *Time Perception, Music, and Immersion in Videogames,* authors Timothy Sanders and Paul Cairns conclude that music can deepen our immersion into a game. Being immersed in this context could be construed as increasing your connection with the game-world and consequently loosening your connection with the physical world. They further conclude that the addition of (enjoyable/fitting) music to a game decreased the player's grasp of time if they were making an active effort to track that time, further implicating some disconnection from the 'real world'. (Sanders, Cairns, 2010)

The authors are careful to point out their research is not definitive, but if we accept that they are at least meaningful, it could mean something interesting for my own work. I don't only make videogame music, but it is a significant aspect of my artistic profile, and when I do make that kind of music I might be able to directly influence my audience's perception of music—or at the very least, I'll be in touch with collaborators with that power.

2.2.1 The Purpose; Not to be Heard?

An argument has been made that music in film is not to be heard at all, but to act as the allegorical typeface to the lettering of our work. From this, doctor Jamie Sexton extrapolates that for film music- and game music alike- you can't simply consider the literal *music* as the entire composition. Instead, we ought take into account all of the sounds, both ambient and SFX, as creating one larger audio-musical score. From this, the theory of *the refrain* is referenced, which positions this type of big-picture audio-music as a sort of entity marking its territory. Imagine a group of birds singing, or a motorcycle racing past, as entities competing for this limited territory. "I can't even hear myself think," being an admission that your territory has, for the moment, been conquered by these sonic entities (Sexton, 2007). Where Sexton and I differ, is that he describes music as being in opposition to the refrain: it deterritorializes sonic space and leaves it open for your habitation. I think this is a popular view, but part of the point of this paper is my outright rejection of music as peaceful cohabitant.

My aim as a musician (emphasis on MY aim- others' may differ) is to shape some part of your perception. Not to bend it to my will, but to, for a moment, take over part of that territory as a foreign entity. I'll be the first to admit I, too, often make music that simply cohabitates. My best works, however, challenge the listener. For some games, this might not be an appropriate attitude. You don't want to be occupied with the music as you try to figure out a tough puzzle, for instance. However, for games where you have a little more mental real-estate, I want to aim at taking some of that and leaving my mark.

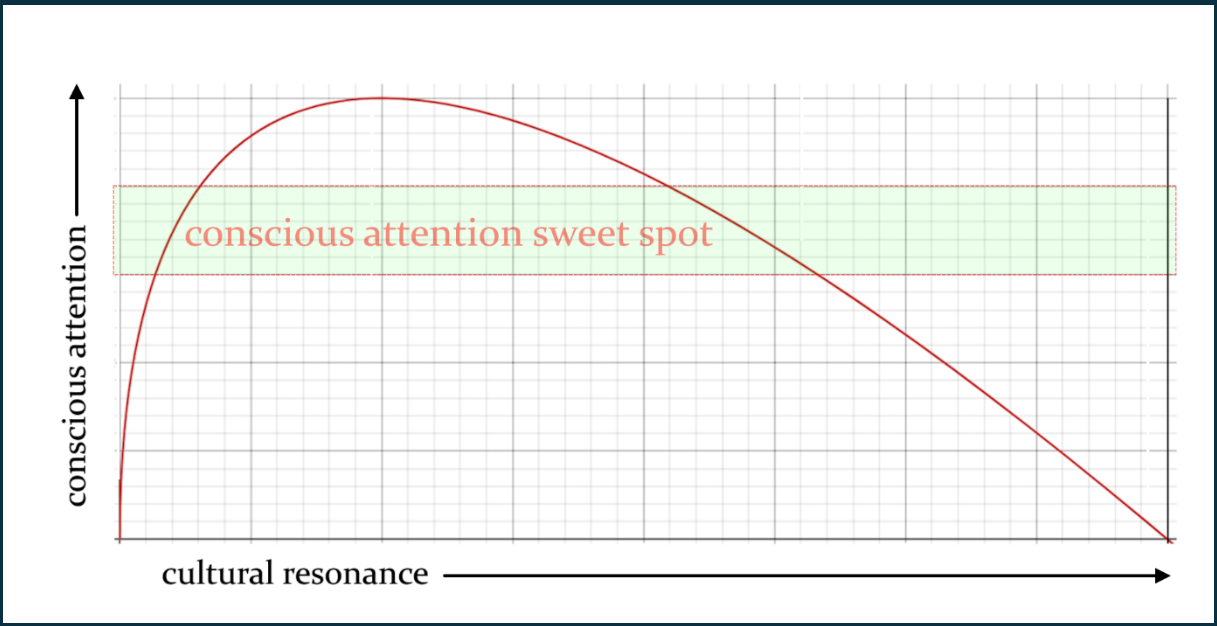
This aim is further complicated by a later passage from Sexton's book:

> "a study commissioned by the BBC in 2005 [...] found that at least 70 per cent of them [gamers] had other media on in the background, either the television, radio or personal music (Pratchett 2005: 5).
> I imagine this finding will make uncomfortable reading for those who study video-game music exclusively within an aesthetic tradition of film music, because it suggests that aesthetic considerations are secondary to music's ability as an aid to concentration. Of course the survey data needs to be qualified further before such an assertion can be made. For instance, it utilized a very broad definition of 'video game', which included the solitaire game bundled with the 'Windows' operating system and phone games."
> (Sexton, 2007)

Even with the noted qualifying factors, we might conclude that music for games isn't capable of being part of this *refrain;* that it's simply a helpful factor to immerse ourselves further without any specific artistic contribution for a majority of our audience. I certainly think this is a challenge, though not one that hasn't been implicitly included in earlier findings in this paper. After all, if sound is hard to hear- or simply switched off- then that is an aspect of ecology. Throwing our hands up in defeat seems like a fairly unambitious response. What I wonder is in what cases gamers *do* listen to the music.

Anecdotally, I've spent a lot of time playing Dark Souls I with the sound off and youtube on. Often this was during a replay, or during a section on which I was stuck for a long time. In other words, when I was either under- or overstimulated by the gameplay. The contrary tends to be achieved by games in which the soundtrack isn't static. By that I mean, there's more than just a loop playing, the music moves *with* you, rather than *around* you. In an ecological framework, this might be explained as music that's attached to the world (enter a level, the music starts. New level, new song. Etc.), or music that's attached to you (enter the game, music starts. New level, new instruments/rhythm/dynamic/etc.) And I think this bears out when put aside the earlier *refrain* theory. After all, if music is part of a (game)world-wide score of sounds and ambiances, it becomes a background of comfortable, space- deterritorialized. Carrying that music with you instead, aware of it's changes as pointed breaks in repetition, forces you to internalize it- to be territorialized.

We might bring this back to the earlier concept of invariants. Considering the earlier attention/resonance graph, you could come to the conclusion that the balance to be found, in this case, would be to cause enough un-resonance (friction) for the listener to be notice, but not so much that the listener becomes "bewildered":



An example that has, anecdotally, hit this sweet spot is Hades. In Hades you attempt to escape from the titular underworld of Grecian legend, accompanied by a fairly typical rock-track. However, an audience unfamiliar with east- and mid-European musical tradition might be caught off guard somewhat by the Grecian bouzouki and the Turkish lavta and baglama, all stringed instruments in the same family as the lute but with very distinctive sounds. This moves us to the left of the resonance axis. We're further moved when the soundtrack, though a constant companion, radically transformed between combat, social encounters and places of respite.

For me personally, this was a combination of sonic curveballs that brought me to a point where all I wanted to do was listen to the music whilst playing- the game would be lessened by listening to something else! And really, this isn't too much to ask of ourselves as composers and musicians. There are so incredibly many musical traditions with which we may be more or less familiar, and *fully* adaptive music is really the gold standard of this field. This might be a lot of work, but it's work we (I) can and should do to make music a more notable part of our games.

2.3 Mario's Mates Make Musically Matching Motions

Mario, the titular character of his series of games, has been a household name the world over since his conception in *Super Mario Bros,* released in 1985 by Nintendo. Nintendo as a company has made a major effort to conquer a videogame market that the other major consoles (the Playstation and the Xbox) have neglected somewhat- namely, families. This puts Nintendo games into a position where there are some assumptions made about the ecology when creating its games, which I'll be discussing momentarily. Due to these ecological assumptions, drawn from Nintendo's own marketing and game designers, we have a clear setting in which to analyze their musical choices.

Since the Mario Bros franchise is fairly expansive, I'm not going to linger long on any particular game, but more on overall trends. I will, however, back up my theories with quotes from the developers and, when applicable, general research.

2.3.1 Mario's Ecology

Having mention the predictable ecology, let's take a moment to more clearly frame it. As I've established, ecology in the context of videogames is not a single set of environmental interactions, but two: the 'real world' interactions with your direct environment, and the interactions with the virtual environment which has been shaped by the creators of the game.

This first environment, the 'real world', is the one we have no direct control over, but which we can influence and become more aware of through market research and conscious game design. Nintendo took a divergent path in 2006 by releasing the Wii, a console created explicitly for families and with motion controls- a type of technology that effectively demands a spacious play space to accommodate it. This shift granted them a unique position in the videogame market, which had gradually shifted to become more focused on adolescent and pre-adolescent boys (Davnall, Z.D.). Furthermore, it allowed them to assume that people would play the games in a communal area, preferably one with space to move around. Though not all games in this series since the release of the wii have been multiplayer-first ventures, it's still a common trope which they fall back on and one which they seemingly emulate in other games in the series simply for the sake of consistency.

2.3.2 Dancing Koopas

A significant choice early choice, and one I would argue is intrinsically tied to ecology, is to have the little enemies on screen stopping to do a little dance-move on specific beats of the music in *New Super Mario Bros.* and *New Super Mario Bros. Wii*. During an interview with the US branch of Nintendo, Kondo expressed that this element of visual/auditory crossover was an early attempt to tie the gameplay to audio (Nintendo USA, 2023). Kondo further shared that there was some frustration with this lack of overlap: he had wanted to do more, but since audio was often brought in late into the development of a game, there was less space to tie musical ideas directly to gameplay.

Despite Kondo's understandable desire to do more, the dancing enemies became a hit with fans, something which you can see mentioned in several fan forums as an undeniably positive aspect of the game. It's odd, then, that this excitement would all stem from such an innocuous seeming element. Yes, the enemies have always been made intentionally appealing in Mario games and yes, a little dance is obviously cute, but I argue it's more than that.

In personal correspondence during my development of this paper, I've had a lot of people referencing Mario games as one of the few categories of games where they keep the sound turned on, and I believe this is very much related to small elements like this that Nintendo keeps intentionally inserting to make the music an integrated element of their worlds. If these elements are what get you to hear the music more consciously, it suddenly becomes a much more obvious and maybe even a much a more significant part of the game, as it's a very solid connector to a whole aspect of a videogame you'd otherwise (more) easily tune out.

You could imagine that playing with two or three rowdy children and maybe a parent, it would be hard to become immersed into the world of NSMBW, much less pay conscious mind to the music. The dance moves help break through that. To quote experimental psychologist Annabel J. Cohen:

> “the temporal structure of an object expanding or contracting three times a second can match a sound that increases and decreases in intensity three times a second. In this example, the formal properties of visual objects are congruent with those of the music. The relation is not necessary, but, it is proposed that when this structural congruence occurs it has special significance to the viewer-listener. This reasoning follows from theories about the significance of formal congruence *within* a modality. For example, Bregman (1993) discusses auditory scene analysis as dependent on properties described by the Gestalt psychologists as similarity, proximity, and good continuation.” (Cohen, 1998)

To be clear: yes, we’re still talking about dancing turtles that dance to a flighty “*Baa, baa!*” it feels a bit silly to take an element like this so seriously. But in truth, making music for games is often enough a silly exercise that we take seriously in spite of ourselves.

Interestingly, the soundfont of *new Super Mario Bros Wii.* is much less saturated than that of *New Super Mario Bros.,* with an emptier high-end. It has been argued that this makes the music less energized (SilokHawk, 2020), though I actually think this strikes a nice balance when combined with the dancing enemies.
To return to the concept of *the refrain*, music which is described as more penetrative is really more *territorial*. When we continue to interpret this music through the lens of local ecology, it makes a lot of sense to have elements that gently nudge someone towards the music than it does to have a territorial piece of music fighting with a rowdy environment of people playing a game together.

The koopas in NSMBW were a precursor of a greater trend in Mario games. In *Super Mario Odyssey*, for instance, the ‘captures’ adjust their sound effects to the current chord of the background music, creating a similar (though distinctly less audio-*visual*) tie from gameplay to music. The place where this is most notable is in the electric cables through which you travel, which emit an energized arpeggio whenever you move through them.

The musical elements of Mario were further elaborated in *Super Mario Bros. Wonder,* where there are dedicated levels which ask you to interact with its music differently. In the level *Ninji Jump Party,* you’re rewarded for jumping in line with the background music’s beat- though most of the level can be finished without engaging with this dynamic at all. I do find it a bit unfortunate to see that this has moved musical interaction from an inherent and popular game element to something isolated to dedicated levels. Yes, KK and his coworkers could do more in these levels, but I’m concerned the average player will actually leave music behind in the rest of the game this way. After all: the music level is over- why continue to pay attention to the music?

3.0 The Experiment

Now that we've grasped the concept of ecological music analysis, it's time to dive into my own work and take a serious look at how all of this applies to my creations. As noted in my introduction, themes discussed in this paper have gradually become a greater priority for me throughout the 2023/2024 academic year. As such, there's a curve of ecological awareness that we can follow, as well as a critical perspective to take regarding the elements of local ecological awareness that are lacking. Both of these approaches are going to be further discussed in this chapter.

3.1 Methodology

To evaluate my use of ecological awareness, I'll be observing my own work through two lenses: An affirmative perspective, which takes into account the ways in which I've increasingly used an imagined perceiver's ecological circumstance in my creations- and a dissenting perspective, which will be a case study of a recent piece in which I worked without a conscious awareness of ecology.
Note that this latter perspective isn't intended as self-deprecation. I'm simply aware that local ecology isn't always on my mind during the creation process, and chose to perform this experiment as a way of comparing what I've learned this year and while writing this SN to the way in which I've worked for a long time and which has left me somewhat dissatisfied.

3.2 Retrospective Analysis

I'll be taking a look at three pieces and discussing my awareness of local ecology in each of them:

Surreality Check is the game my company has spent the last two years building and for which I've written a lot of music *before* starting my research into this subject. Its music uses a lot of motifs and even some in-game ecological awareness, but is still a good baseline to see where I've come from.

KASKO / Kampen was a music-theater retreat during which me and a group of fellow artists created a set of music pieces throughout a the spaces of an old church. You might remember from the introduction that this experience put me onto this subject in the first place, and although I have an unfortunate lack of recordings of the experience I took rigorous notes during and am confident it'll be a useful place to analyze my start into ecological consideration.

Cogs is my final piece of ecologically-centered music of the year. This is another piece I mentioned in passing in my introduction. I wrote it for the Zuilens Fanfare Orchestra, and it was unfortunately delayed due to its challenge for the performers. Though this means I'll not be able to compare the imagined ecology to the "true" ecology, I still believe it's a useful case study of how I'm currently putting these ideas into practice.

3.2.1 Surreality Check

After two years of development, my company is getting perilously close to finally releasing our debut title. In Surreality Check you play as Rowan Roach, an insecure office worker who gets sent to company mandated therapy after a series of mental breakdowns. Instead of a therapist, however, he ends up in contact with a series of odd manifestations of his subconscious as well an entity claiming to be The Universe itself. The 3d puzzler / graphical novel lures interested players into an introductory exploration of absurdist philosophy and behavioral psychology with an art style that can best be described as fever dream with office overtones.

This isn't a game where ecology was on my mind very much, as I started working on it well before I even started my final year at the HKU. In spite of this I can still see reflections of this topic already present in my work.

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The main menu theme consists of a mysterious sounding set of synthesizers, with an extremely close by main line and a couple of very far, twinkly synthesizers. As I said before, none of this was a conscious attempt to reflect any kind of ecology, but when I put it together with the background of the main menu it does still click together:

If we consider this as a virtual ecology, I definitely made some subconscious ties to it to draw a player in. The low, close by synth sounds how the cramped office on the right looks, but to your left the starry sky opens up, and in the background a reverb-drenched trinkle confirms this other side of the image as well.

All of this merges with the final fact of the score: the entire way through, you hear a voice humming along. You even get the opportunity to turn it off after a while, which it does with one of a number of randomly chosen apologetic remarks ("whoops, my bad"). This is a stand-in for roach, and is intended to immediately put you into his shoes as a slightly awkward, socially anxious person who's fallen just slightly outside of the 'normal' label.

For someone not thinking consciously about ecology, there's already a clear interest here to make something happen! This really coincides with my growing unease at the time. An early attempt to connect more with my audience- to get a better sense of how my music would sound to their ears- I had started to include these bits of non-singing vocals to my music. The way that combines with the environment makes me think a lot of the "baa-baa!"s from *Super Mario Bros. Wii,* in how it pulls a human element into the music to grab your attention.

This is all still very metaphorical. I get more literal (and, perhaps, more predictable) when I go into an actual level.

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The lobby theme is really one central theme, with three modular layers:

you enter the lobby to start the game proper, and get the opportunity to snoop around a bit as you look for ways to distract a minor antagonist. As you do, the central theme reflects the main theme with some intrusive synths, though it adds a supporting piano layer which helps define its 'central'-ness.
When you find a distraction and go into a sideroom, however, an appropriate layer is added based on the character you find inside and the central theme is filtered through a lowpass. The effect is that it sounds like stepping into a different space but always hearing the central room through the door.
This is, again, a kind of ecological thinking, though I kind of dislike its literality.

Considering the earlier *refrain* theory as it interacts with the attention/resonance graph, there might be something to keeping the central theme simpler to let it live outside of the *refrain* and below our conscious attention sweet spot. That way, you could ratchet up the tension once we enter a sideroom from which we're explicitly forbidden. Here, adding differently contrasting elements depending on the character, we can pull a player into the aforementioned sweet spot, which helps them notice the more specific musical character specifiers- rather than just being another piece of music.

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The Doctor Uni theme is the piece of music which a) is most in need of being replaced and b) is the most central to the message of the game. Here, after you've messed around in the lobby, you finally come into contact with the entity that's supposed to be helping you get better… And it's an absolute prick. It belittles you, dismisses your problems as insignificant, makes you feel small and unimportant and generally treats you with a kind of disrespect completely unbecoming of a doctor of psychology—then again, why assume it is what it says it is?

The music here is there least thematically and ecologically sound. A teacher of mine once described it as "old man jazz", and it's really a placeholder track made towards the start of the game. In this beautifully designed, atmospheric room, during the climax of each level… You're in a shopping mall which plays stock music over the speakers because it didn't want to make a deal with a radio-station.

This is where the intro music should be coming back in. As you may have guessed, the picture I showed earlier is of Uni's office, and the intro music already resonates with this environment effectively. The risk discussed in the attention/resonance graph comes to fruition here: the invoked invariants are so completely disjointed from the context that a player is left bewildered.

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The real-life ecology of Surreality Check only came into view as I researched this paper. Truthfully, our core demographic hadn't even been very clear before I started working on my supportive narrative. During this research, we ran a playtest to get a more concrete idea of what our direction ought to be. As it turns out, this type of psychology/philosophy 101 is very attractive to adolescents and some students in their early 20's, usually introverted, most likely PC gamers. This says a lot about ecology! So- in what context will someone be playing this game and hearing my music? To answer this question I've made a brief ecological profile- inspired by the more typical 'client profile' approach.

> Sophie is a 19 year old student. She just graduated high school and is now going to collage near her home town. She picked her collage together with her best friend so they wouldn't start alone in a new school. Sophie doesn't game more than a couple of hours a week, and has a preference for story-driven games with little emphasis on tests of reflexes. She likes to lose herself in a world where she feels safe and in control. When she does play, she plays in her room on her PC, with her door closed and her headphones on. When she plays games she's often played before, she might turn on her own music and only leave the game's SFX on, though for new games she tries to take in the whole experience at least once.

Hypotheticals are, of course, limited, even when they're based on playtest data. That said, this is a very clear ecology that I *could* incorporate before release! For instance:

- Playing with your headphones on is common among people who play explicitly for escapism. A consequence is that the music is very close to you, and that little textural details are less likely to be lost. The twinkling sounds from my intro music might be something I want to bring back more often- or maybe I can think of different small textures, like a more nervous character periodically shuffling their feet around in the rhythm of the music, or a creaking chair softly adding to the ambiance of a quiet bit of atmosphere.
- Since players who play on their own in a private space don't tend to have as many distractions around them, it could be that the music needs to be a little more stimulating to keep them from simply turning on their own thing: it needs to have enough elements to keep someone from getting bored, and not be too short, to occupy the sonic territory that would otherwise be takes up by said distractions
- Sophie plays to experience safe social contact. The music can support this by emphasizing characters' personality traits. This is something the current side rooms already do really well! It could maybe be enhanced by allowing the characters to more directly interact with the music, like the foot shuffle mentioned before. Rhythmic feet shuffling might be a bit weird and embarrassing in the real-world, but here it's a confirmation that there are no behaviors that will lead to ostracization.

Incorporating these elements could help the experience of players like Sophie.

3.2.2 Kasko

On November 20th of 2023 I stepped into a car and drove to Kampen, Overijssel. I stayed here until the 24th, in a small vacation home just outside of town with 5 other students. Every morning we made our way to the Broederkerk, where we prepared six pieces of theater music, each for a different space in the church. It was an exhausting, intimidating experience that resulted in a beautiful anthology of 5 minute pieces, each with their translation of the space in which they were performed.

To me, Kampen is where this supportive narrative came to life. Ecology was an unavoidable aspect- but it was also because I don't often perform live, much less in such proximity to an audience, and it helped me realize what was missing from my music previously. For this analysis I'd have loved to go through each performance, but due to the word limit I'll illustrate what appealed to me by describing the first two acts.

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The audience enters through a narrow brick hallway with a transparent plastic ceiling. It's cold, damp, and cramped. From the ceiling hang a dozen or so low quality headphones, quietly playing recorded fragments from repetitions of the later pieces. The audience occupies themselves with the different sets of headphones, shuffling past each other with a periodic whispered 'excuse me'.

This piece is probably the purest embodiment of ecology as performance. The environment is shaped in such a way that interaction with the other audience members in it is unavoidable if you want to experience the act, and the barely audible whispers and music from the headphones implore you to squeeze past the cold walls and reach up to the somewhat leaky ceiling. This is where I want to emphasize that ecology isn't *just* environment, but the interaction *with* your environment. Performing in this hall is not the same as exploring the hall to be able to *reach* the performance.

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As the doors at the end of the hall open, the audience streams into the next room for **the second act.** This second room is the same width as the hall, though it's significantly shorter and leaves no room for movement once filled up with audience. Though still damp, this room is significantly warmer than the hall and has a bit of a musty smell. As the audience enters, the pianist in the corner produces a quiet ambience; a muted tremolo of D2 and D3. The ambiance builds as the vocalist- positioned crouched down in a closet and invisible to the audience, though audible through two large speakers- starts to mumble like a hungover sailor. The piano follows this mumbling as it slowly outlines a chord progression in G harmonic minor.

Throughout the act, the vocalist moves out of the closet and into the room with the audience, stumbling across the walls as she sings a song written for the performance with a saxophonist and the pianist accompanying her. As she reaches the audience, the song reaches its climax with a sudden incursion of dubstep which blends into the rest of the arrangement. At the height of this climax each of the other performers takes a moment to observe the singer, only to leave the room as her over-the-top demeanor takes up more and more space, pushing the audience into the spaces which were previously reserved by the other performers.

By the end, the singer is laying on the floor blurting out lyrics in French. The audience leaves through a side door, walking around her and leaving her alone as the door closes behind the final audience member.

This is a super weird, super cool piece and a ton of fun to perform. The interactions with the absurd character of the singer were made especially uncomfortable by the cramped space, but in exactly the intended way. There's a sort of ebbing to the way in which the audience moves around the space- from nervously waiting, to swaying to the music and moving around the main character, and finally being pushed away and to the sides. This ebbing elevates the music from *music* to *performance*. Furthermore, the resonance/attention factor is off the charts here: bewilderment is desired! This is a context in which a little church sideroom, sometimes used for displaying the recently deceased, is filled with a drunken chanson and dubstep—the invariants are all over the place!

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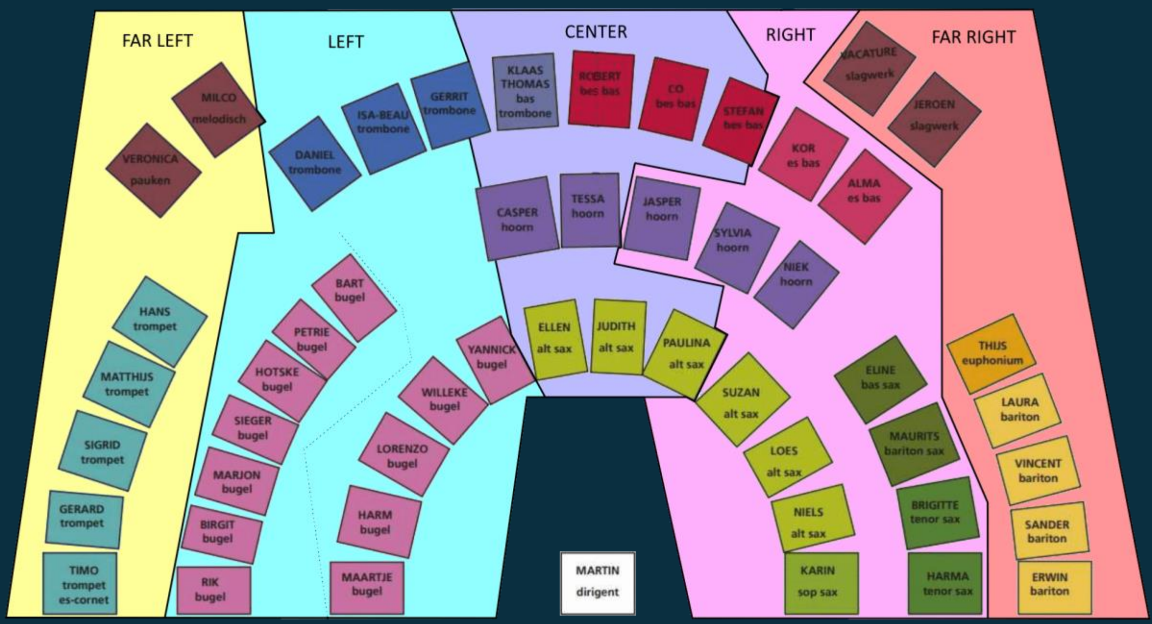
Kampen was a wonderful dive into the way in which an audience's experience of space and environments can be a shaping element of the artistic creation process. It's also a wonderful example of using ecological factors to distort and reimagine established invariants through recontextualization. In my next piece, I decided to try to consciously use this shaping to enhance the experience of my composition.

3.2.3 Cogs

Nearing the year's end, I got the opportunity to have my music played by a fanfare. I spoke with the band members, the conductor, my teachers, and explored the location of the performance- all to get a grip of how this performance would look and feel like once it was performed live. In the end, I decided to place my main focus on directionality. That is to say, I wanted to have a sort of live-panning be central to the music. A fanfare is composed of such a great number of musicians, after all, that a bit of coordination could turn this one piece of music into a showcase of the sheer scope of cooperation I was experiencing.

The first task at hand was to make a distribution of players. I downloaded the band's seating chart and made a distribution of directionality:



This way, I could keep track in my composition of not just *what* I was writing, but *where* I was writing it. I made a further distribution which I also handed out to the players, designating where they were to be seated- one part was written for 6 (left) bugels, whilst another was written for 4 (right) alto-saxophonists. This to set them apart from the 5 (right) bugels and 3 (left) alto-saxophonists. The vibraphone, a central part of the performance, was moved to be located at the back but in the center lane.

A request I had, which unfortunately could not be fulfilled, was to have the fanfare separated into three columns, through which the audience would have to walk to get to their seats as the performers played an aleatoric introduction. This would've introduced the concept of directionality, whilst also allowing listeners to more actively engage with the artistic environment I was building- thus elaborating on the ecological aspect.

The piece itself took its time to slowly introduce the aspect of directionality in a couple of phases:

1. The vibraphone plays a series of repeating patterns. It's in the center, playing on its own.
2. The other percussion instruments add their own elements from the flanks. Their sonic territory is not quite as large as any section of brass instruments, so there's still a lot of space left over.
3. The brass instruments come to life, but only one side at a time. Each chord change flips from a combination of FL, L, C to C, R, FR. This way the implied directionality of the percussion is confirmed and reinforced during a little over a minute of iterations on the harmonic material.
4. An increasingly intrusive timpani indicates the coming of the climax.
5. Overlapping melodies shoot back and forward. For instance, a melodic line might be split between even notes played by the tenor saxophones and the uneven notes played by the (right) bugels.

The full score is included for further clarification. By this way of writing, an attempt is made to make manifest the ways in which this orchestra is a large, 'multicellular organism'. From experiencing the repetitions, I can whole-heartedly confirm that this idea comes across. The tale does have a bittersweet ending: due to the difficulty in playing music that requires this level of cross-band coordination, the piece couldn't be performed when planned. It is currently scheduled for spring 2025.

3.3 Experiment: Alone

As the fanfare collaboration came to an end (or in my case: took a sabbatical) I decided on creating one final piece to look into confirming some things I'd theorized about in this paper, but had yet to confirm. To that end, I finished a project that had been laying on the shelf for some time, and spread it around alongside a survey with questions about people's setting and enjoyment. The project in question, tentatively titled 'Alone', hadn't been created with the knowledge of local ecology I'd been gathering this year, and in finishing it I made a conscious effort to try to keep it from being too ecologically 'aware'. I know how I feel ecological awareness could bring me closer to my audience, but this experiment was really intended to get a better grasp on another important aspect of this research:

Does it matter to the player?

Lacking satisfaction is enough reason to start this research, but a lot of it has ended up being about the way in which it might improve a listener/player/perceiver's experience of my art. To do so, it would be important to check if the things I worried about were relevant in the first place.

3.3.1 The Game and Survey

Alone is a simple puzzle-platformer. You jump around a level to get to the upper right exit, at which point you reach the following level. The music is a relatively simple 4-track bittune consisting of a melody line, a counter melody line, a bass and a noise-envelope percussion—the sound effects are all created using the same VST synth as the music.

During the creation of Alone, physical ecology wasn't considered, and virtual ecology was only a minor consideration. The one major aspect that could be put into this bucket is related to the twist of the game: after every 5th level you're returned to level 1. When this happens, you lose one of the blocks which form your character, decreasing your size, but also removing your option to move into the corresponding direction that block represented. Each time this happens, another track of the music is stripped, until you have only 1 block left which can go up, nor right, nor left, and the melody, similarly stripped of its supporting elements, continues indefinitely as the title of the game fades in.

The survey that accompanied the game (enclosed in this paper's folder) asked about ecological factors; where did you play the game?
Was your environment distracting?
Was your environment noisy?
etc.

... about the ways in which the user engaged with the game;
Did you change any of the settings?
Did you listen to other media whilst playing?
Through what kind of device did you play the audio?

... and finally, about the quality of their experience;
Would different music have lessened your experience?
Did you enjoy your time with the game?
Optionally: did you hear the music over your environment, if it was noisy?

3.3.2 The Results

After giving the survey some time to gather responses, I took the time to review what I'd learned. Below I've registered my most significant findings. None of these findings are conclusive. Rather, each indicates a correlation which merits further exploration in future. Playtests come and go, and for these topics in particular I plan to include more questions in future to try to get to the bottom of exactly what is happening.

Distractions correlate with decreased interest in music.

Perhaps obvious, but still important to note, is players with more distracting stimuli in their environment and players with a louder environment less often felt that the game experience would've been lessened by different music. Though predictable, it's still good for the experiment to confirm that environmental distractions indeed lessen the enjoyment of the music- or at least the perceived necessity of the music for the game quality. There could be a question as to whether this correlates to the relative simplicity of this track, or if this would be true for all music.
In future, I might ask more about distractions and see if I can't figure out if *territorializing* elements in music have a more negative effect in players who played in a more distracting environment.

Spacious environments inversely correlate to satisfaction with the sound effects.

Less obvious, though encouraging, is that players who played in a more spacious environment were less satisfied with the game's sound effects. I call this encouraging in the sense that it seems to agree with the hypothetical mentioned in passing earlier in this paper, about how the dalmation image might read as a large map to someone watching it in a busy mall: it seems that the sonic sense of scale is subconsciously expected to grow alongside the environment. This alone could shape an entire follow-up study and this is not nearly enough data to be conclusive, but it could be an interesting question to utilize in future playtests.

The number of sittings inversely correlate with the muting of sound effects and music.

I included a question about whether or not SFX and/or music were muted during play, as a follow-up to the earlier discussion of said topic. I think the results here might be biased. I tried to not emphasize that my thesis was about music and technology, but it still happened to come up semi-regularly. In those cases you could imagine a player might be more likely to keep music on, just to be helpful. It would at least go a long way to explain this result- if you played in multiple sittings instead of skipping through, you might have more motivation to play in what you presume is the *intended* way. If this *doesn't* explain the results, I'm quite baffled as to what would. My expectation would be that an immersed player would play in 1 sitting (especially since the game isn't all that long) and would keep the music on, but instead the more often you played the less likely you were to turn off music.
It could be that the causation is backward: if the music is irritating, that could cause someone who doesn't turn off the music to play in multiple sittings rather than continuing to listen to the

music whilst playing the game in one go. This is another topic that more specific questions might clear up in future.

4.0 Conclusion

I started this supportive narrative with an unsettled feeling. It seemed that something was missing from my compositional process which I could not quite grasp. Once I started searching, that feeling turned into a curiosity: I was going to explore how a listener interacts with their environment and how I could influence that interaction to place my music in as favorable a position as possible to be experienced.

At the end of this process, I can say with certainty that learning about the ecology of my audience is a useful way of creating music that connects with them- and that makes *me* feel more connected as well.

Important findings include the concept of ***Invariants;*** parts of a whole that define that whole in our mind, and on which we rely to make sense of the world. Related this this is the ***Resonance/Attention Curve;*** an attempt I made to estimate in which way using invariants in unexpected ways could draw in the attention of- or risk alienating- your audience.

I also applied this concept to the ***Refrain Theory;*** which conceptualizes our capacity of sound perception as a territorial struggle, with sound as a conquering force and music having the power to *deterritorialize* our mind. I've argued that we could instead use the displacement of invariants to use music in this same territorializing way, to cause an audience to more consciously hear our music. I've also come to define a ***Resonance/Attention Sweet Spot***, outside of which we risk either losing our audience's attention by over- or underwhelming them, or overstimulating their senses and becoming more distraction than art.

I conceptualized videogame music as existing in a ***Virtual Ecology;*** a place outside the real world which nonetheless becomes real to an audience, and with which music need to interact in a certain way to decrease the odds of becoming superfluous to an experience and being turned off- mainly by ***Attaching to a Player, Not to the World.***

I've analyzed a small sliver of Mario music (I would've liked to discuss it more- alas, 10k word limit), and concluded that ***Synchronized Music/Game Elements*** for an important part of what keep people tuned in to game music.

I compared the progression of my own work to all the above theorizing, and analyzed how I used ecology as I became more aware of it, whilst also suggesting changes to make in future to more actively use ecology. In this process, I also made an ***Ecological Profile;*** a derivative of the popular 'client profile' which specifically attempts to outline the environment in which a client plays and how it interacts with my art.

Finally, I conducted a survey to confirm that my old way of working had an impact on the enjoyment of specifically my game music. The results were interesting, though not conclusive. In future, I'd like to apply my proposed changes to Surreality Check and test whether those improve audio-consciousness and overall game enjoyment.

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